

ELK-Em: Closure-Aware Embeddings for $\mathcal{EL}^{++}$ Ontologies (Technical Report)

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Abstract. $\mathcal{EL}^{++}$ is a lightweight description logic designed to capture the expressive requirements of large practical ontologies while supporting tractable reasoning, as implemented by reasoners such as ELK. Recent efforts have developed scalable methods for learning continuous representations of $\mathcal{EL}^{++}$ ontologies for inductive inference, but they primarily embed explicit axioms, provide no formal guarantees for deductive-closure approximation, and cannot predict links for individuals without ABox assertions. We present ELK-Em, a novel geometric ontology embedding model that improves closure approximation while placing individuals based on lexical similarity. ELK-Em embeds (i) concepts as axis-aligned boxes; (ii) roles with a new parametrisation that supports full $\mathcal{EL}^{++}$ expressivity, enabling soundness and ELK-closure approximation guarantees; and (iii) individuals as points regularised using domain-specific encoders. On GO, GALEN, and Anatomy, ELK-Em leads to an important improvement in the approximation of the ELK-derived deductive closure over geometric baselines. ELK-Em also supports inductive concept-assertion prediction, for example, on human protein-function data using sequence embeddings.

Keywords: Ontology embeddings · $\mathcal{EL}$ Embeddings · biomedical applications

1 Introduction

$\mathcal{EL}^{++}$ is the lightweight description logic underlying OWL 2 EL [18], a profile of the Web Ontology Language [2]. It is designed for ontologies with large numbers of classes and relations, capturing the expressivity needed for such ontologies while retaining polynomial-time reasoning for core reasoning tasks. This makes it suitable for large biomedical ontologies such as the Gene Ontology (GO) [1], SNOMED CT [10], and GALEN [21]. In $\mathcal{EL}^{++}$, domain knowledge is expressed through logical descriptions, or axioms. These axioms are divided into an ABox, containing instance-level assertions such as *Diabetes*(*Alice*) or *hasFinding*(*Alice*, *HighBloodGlucose*), and a TBox, containing concept-level axioms such as *Diabetes* $\sqsubseteq$ *Disease* or *Diabetes* $\sqsubseteq \exists \text{hasFinding}. \text{HighBloodGlucose}$.

The polynomial-time reasoning guarantees of $\mathcal{EL}^{++}$ enable scalable reasoners such as ELK [13] to perform core reasoning tasks, including consistency checking, class expression subsumption, and instance checking, over large ontologies. For example, the

axioms above entail $Disease(Alice)$, which ELK can derive automatically. This enables applications to use both explicit and implicit knowledge under the $\mathcal{EL}^{++}$ semantics. However, such reasoning remains limited to the symbolic input: in the absence of ABox assertions for an individual, a reasoner cannot derive its class membership, even if the individual is associated with annotation assertions, such as labels, comments or external descriptions.

Ontology embedding methods provide one way to support inductive inference over OWL ontologies [7]. Methods such as Onto2Vec [24], OPA2Vec [25], and OWL2Vec* [6] learn vector representations from ontology-derived corpora, including graph walks, axioms, and lexical metadata such as labels, definitions, synonyms, and annotations. These representations can be used for downstream TBox and ABox prediction tasks. However, such methods primarily treat ontologies as graph- or corpus-based inputs and do not explicitly impose description-logic semantics as constraints during training to model the logical structure explicitly. More recent works, [14,17,20,28,11,29], developed geometric embedding models for $\mathcal{EL}^{++}$ ontologies, where concepts, roles, and individuals are represented as geometric objects and axioms are encoded as constraints in the embedding space. These models provide a closer connection between embeddings and description-logic reasoning, but still suffer from the following limitations:

- Existing geometric ontology models learn individual embeddings only from observed ABox facts. They therefore cannot place a new individual with no ABox assertions, even if it has labels, comments, annotations or external domain-specific encoding. This limits applications such as CAFA [9], SNOMED CT entity linking [8], and phenotype annotation [19,5].
- The geometric designs may impose unfair semantic restrictions. For example, as [4] highlights, models might approximate $\exists r.C \sqsubseteq \perp$ as $C \sqsubseteq \perp$, which is not a valid inference. This incorrectly penalises all candidate assertions $C(a)$.
- Methods that embed only a fragment of $\mathcal{EL}^{++}$ cannot preserve all consequences computed by ELK. Since ABox scores depend on the learned TBox geometry, missing TBox consequences may lead to poorly scored assertions for individuals.

To address these limitations, we present ELK-Em, a geometric ontology embedding model that improves approximation of the ELK-closure while placing individuals based on lexical and domain-specific similarity. ELK-Em represents concepts as axis-aligned boxes, roles through a parametrisation designed for $\mathcal{EL}^{++}$ axioms, and individuals as points regularised using domain-specific encoders. We use boxes for concepts following prior work showing that they are better suited than balls for preserving closure under intersection [20]. The role parametrisation allows us to give formal guarantees at zero-loss: (i) soundness, *i.e.*, if ELK-Em achieves zero-loss, then the learned embedding is a model of the ontology in the model-theoretic sense, and (ii) it preserves a large fragment of the ELK closure. On GALEN [21], GO [1] and Anatomy [23], ELK-Em outperforms geometric baselines in approximating the ELK-derived deductive closure. Using ESM-2 embeddings to derive protein-sequence similarities [15], ELK-Em also supports inductive concept-assertion prediction on CAFA5-derived human protein-function data [9].

The rest of the paper is organised as follows. Section 2 reviews related work on ontology embeddings, with a focus on prior geometric embedding methods for $\mathcal{EL}^{++}$.

Section 3 introduces the required background on the $\mathcal{EL}^{++}$ description logic. Section 4 presents ELK-Em, including its geometric semantics, training method, theoretical properties, and lexical regularisation mechanism. Section 5 evaluates the model on deductive-closure approximation and inductive ABox inference. Finally, Section 6 concludes the paper and presents directions for future work.

2 Related Work

Knowledge graph embedding methods preceded much OWL ontology embedding work and inspired representation learning for structured knowledge [27]. They embed RDF-style multi-relational graphs for link prediction; for example, TransE makes $\mathbf{h} + \mathbf{r}$ close to $\mathbf{t}$ for observed triples and distant for corrupted ones [3]. Such methods mainly target ABox-level facts, whereas OWL ontologies also contain TBox axioms with class hierarchies, logical constructors, existential restrictions, and role inclusions. This requires embedding methods that go beyond assertional graph completion and account for OWL entities, axioms, and logical structure.

Ontology embedding methods address this setting [7]. Onto2Vec [24] first expands an OWL ontology with inferred axioms using a reasoner, then serialises the resulting axioms into a text-like corpus for Word2Vec training. OPA2Vec [25] extends this approach by augmenting the corpus with annotation axioms and lexical metadata such as labels, descriptions, and synonyms. OWL2Vec* [6] combines random walks over the ontology graph with lexical information and logical constructors. These methods have been applied to prediction tasks over OWL ontologies, including biomedical ontologies. However, their objectives do not enforce adherence to $\mathcal{EL}^{++}$ semantics in the learned representation; axioms and logical constructors are instead used as corpus or graph-derived context. Moreover, ontology-internal lexical metadata is not always the most informative input for measuring similarity between individuals. In protein-function prediction, amino-acid sequences provide both a domain-specific basis for measuring protein similarity and a central signal for inferring protein function, which motivates CAFA’s evaluation of sequence-based computational annotation methods; ESM-2 embeddings provide sequence-derived representations for this purpose [9,15].

A second line of work develops geometric models for $\mathcal{EL}^{++}$ ontologies. ELEm uses n -balls for concepts and translations for roles [14], and EmEL++ extends this design to cover additional $\mathcal{EL}^{++}$ constructs [17]. ELBE represents concepts as axis-aligned boxes and roles as translations, using boxes to better support intersections [20]. BoxEL also represents concepts as boxes, but models roles using affine transformations to capture more general relational structure [28]. More recent models, including Box²EL and TransBox, further enrich the representation of roles and complex concepts [11,29]. These models are closer to the model-theoretic structure of $\mathcal{EL}^{++}$ than corpus-based embeddings, but they do not establish guarantees with respect to the ELK inference calculus used in this paper, and individual placement remains tied mainly to observed ABox facts rather than external encoders. ELK-Em is motivated by this gap: it combines geometric modelling of $\mathcal{EL}^{++}$ axioms with formal guarantees with respect to ELK inference and encoder-based placement of individuals without relying solely on observed ABox assertions.

3 Background: Description Logic $\mathcal{EL}^{++}$

We consider the tractable Description Logic $\mathcal{EL}^{++}$, a member of the $\mathcal{EL}$ family closely related to the logical core of the OWL 2 EL profile [18] and widely used in large biomedical ontologies [2].

An $\mathcal{EL}^{++}$ signature consists of a set N_C of *concept names*, a set N_R of *role names*, and a set N_I of *individual names*. Concept descriptions are defined recursively using the constructors shown in Table 1. An $\mathcal{EL}^{++}$ ontology $\mathcal{O}$ consists of a TBox $\mathcal{T}$ and an ABox $\mathcal{A}$. The TBox is a finite set of *general concept inclusions* (GCIs), *role inclusions* (RIs), *domain restrictions* (DRs), and *range restrictions* (RRs), whose forms are shown in Table 1. The ABox is a finite set of *concept assertions* and *role assertions*. The members of $\mathcal{T}$ and $\mathcal{A}$ are called *axioms*.

The semantics of concepts and axioms are given by interpretations. An *interpretation* $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ consists of a non-empty set $\Delta^{\mathcal{I}}$, called the *domain*, together with an interpretation function $\cdot^{\mathcal{I}}$ that maps each individual $a \in N_I$ to an element $a^{\mathcal{I}} \in \Delta^{\mathcal{I}}$, each concept name $A \in N_C$ to a set $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$, and each role name $r \in N_R$ to a binary relation $r^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$. This interpretation function is extended to complex concept descriptions and axioms as shown in Table 1. An interpretation $\mathcal{I}$ *satisfies* an axiom α , written $\mathcal{I} \models \alpha$, if it satisfies the corresponding condition in Table 1. $\mathcal{I}$ is model of an ontology $\mathcal{O} = \mathcal{T} \cup \mathcal{A}$, written $\mathcal{I} \models \mathcal{O}$, if it satisfies every axiom in $\mathcal{T}$ and $\mathcal{A}$. An ontology $\mathcal{O}$ is *satisfiable* if it has a model. An axiom α is *entailed* by $\mathcal{O}$, written $\mathcal{O} \models \alpha$, if every model $\mathcal{I}$ of $\mathcal{O}$ satisfies α . The *deductive closure* of $\mathcal{O}$ is the set of all axioms entailed by $\mathcal{O}$.

Example 1. The following example ontology describes a family domain:

$$\begin{aligned} \mathcal{T} = \{ & \text{Father} \sqsubseteq \text{Parent} \sqcap \text{Male}, \text{Mother} \sqsubseteq \text{Parent} \sqcap \text{Female}, \\ & \text{Child} \sqsubseteq \exists \text{hasParent.Father}, \text{Child} \sqsubseteq \exists \text{hasParent.Mother}, \\ & \text{hasParent} \sqsubseteq \text{relatedTo} \}, \\ \mathcal{A} = \{ & \text{Father}(\text{Alex}), \text{Child}(\text{Bob}), \text{hasParent}(\text{Bob}, \text{Alex}) \}. \end{aligned}$$

Table 1. Syntax and semantics of $\mathcal{EL}^{++}$ constructors and axioms

Name	Syntax	Semantics
Top	$\top$	$\Delta^{\mathcal{I}}$
Bottom	$\perp$	$\emptyset$
Nominals	$\{a\}$	$\{a^{\mathcal{I}}\}$
Conjunction	$C \sqcap D$	$C^{\mathcal{I}} \cap D^{\mathcal{I}}$
Existential Rest.	$\exists r.C$	$\{x \in \Delta^{\mathcal{I}} \mid \exists y \in C^{\mathcal{I}}, (x, y) \in r^{\mathcal{I}}\}$
Gen. CI	$C \sqsubseteq D$	$C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$
Domain Rest.	$\text{dom}(r) \sqsubseteq C$	$r^{\mathcal{I}} \subseteq C^{\mathcal{I}} \times \Delta^{\mathcal{I}}$
Range Rest.	$\text{ran}(r) \sqsubseteq C$	$r^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times C^{\mathcal{I}}$
Role Inc.	$r_1 \circ \dots \circ r_k \sqsubseteq s$	$r_1^{\mathcal{I}} \circ \dots \circ r_k^{\mathcal{I}} \subseteq s^{\mathcal{I}}$
Conc. Assertion	$A(a)$	$a^{\mathcal{I}} \in A^{\mathcal{I}}$
Role Assertion	$r(a, b)$	$(a^{\mathcal{I}}, b^{\mathcal{I}}) \in r^{\mathcal{I}}$

$$\begin{array}{ll}
R_0 & \frac{}{C \sqsubseteq C} \\
R_{\perp} & \frac{E \sqsubseteq \exists R.C \quad C \sqsubseteq \perp}{E \sqsubseteq \perp} \\
R_{\sqcap}^+ & \frac{C \sqsubseteq D_1 \quad C \sqsubseteq D_2}{C \sqsubseteq D_1 \sqcap D_2} \\
R_{\sqsubseteq} & \frac{C \sqsubseteq D}{C \sqsubseteq E} : D \sqsubseteq E \in \mathcal{O} \\
R_{\top} & \frac{}{C \sqsubseteq \top} \\
R_{\sqcap}^- & \frac{C \sqsubseteq D_1 \sqcap D_2}{C \sqsubseteq D_1 \quad C \sqsubseteq D_2} \\
R_{\exists} & \frac{E \sqsubseteq \exists R.C \quad C \sqsubseteq D}{E \sqsubseteq \exists R.D} \\
R_H & \frac{E \sqsubseteq \exists R.C}{E \sqsubseteq \exists S.C} : R \sqsubseteq S \in \mathcal{O} \\
R_{\circ} & \frac{E \sqsubseteq \exists R_1.C \quad C \sqsubseteq \exists R_2.D}{E \sqsubseteq \exists S.D} : R_1 \circ R_2 \sqsubseteq S \in \mathcal{O}
\end{array}$$

Fig. 1. Basic inference rules for reasoning in $\mathcal{EL}^{++}$, following [13]. Expressions above the line are (derived) premises, expressions below the line are derived conclusions and expressions after the colon are side conditions, *i.e.*, fixed axioms of the input ontology that fire the corresponding rules application. Side conditions can be read as additional premises for simplicity.

A *normalised* $\mathcal{EL}^{++}$ ontology contains only axioms of the forms $C_1 \sqsubseteq D$, $C_1 \sqcap C_2 \sqsubseteq D$, $C_1 \sqsubseteq \exists r.C_2$, $\exists r.C_1 \sqsubseteq D$, $r \sqsubseteq s$, and $r_1 \circ r_2 \sqsubseteq s$, where C_1, C_2, D are named concepts, D may be $\perp$, and r, s, r_1, r_2 are named roles [2]. These correspond to normal forms NF1–NF6, respectively. Any $\mathcal{EL}^{++}$ ontology $\mathcal{O}$ can be turned into a normalised ontology $\mathcal{O}'$ that is a *conservative extension* of $\mathcal{O}$, *i.e.*, every model of $\mathcal{O}'$ is also a model of $\mathcal{O}$ and every model of $\mathcal{O}$ can be extended to a model of $\mathcal{O}'$ [16]. This procedure can be executed in linear time and yields a normalised ontology whose size is linear in the size of the original ontology.

ELK [13] is an $\mathcal{EL}^{++}$ reasoner used for ontology classification. Classification computes the entailed subsumption hierarchy between named concepts, *i.e.*, inclusions of the form $A \sqsubseteq B$ where A and B are named concepts. ELK’s classification procedure is based on the sound and complete inference rules for $\mathcal{EL}^{++}$ summarised in Fig. 1.

4 Method

In this section, we introduce the ELK-Em model. Figure 2 depicts an overview of the main architectural components of ELK-Em. ELK-Em accepts as input any OWL ontology, but only preserves $\mathcal{EL}^{++}$ TBox axioms, which need to be normalised before computing the geometric embeddings. ELK-Em also encodes ABox axioms that may be given together with the TBox or in independent files. Additionally, ELK-Em can also combine the geometric embedding with lexical embedding (if available) to improve the placement of unseen instances. In this paper, we apply ELK-Em embedding to approximate the ELK-closure and to perform class assertion predictions.

The remainder of this section presents the geometric design of ELK-Em, its training procedure based on differentiable loss functions, the proposed scoring method, the associated formal guarantees, and the individual-level lexical regularisation strategy.

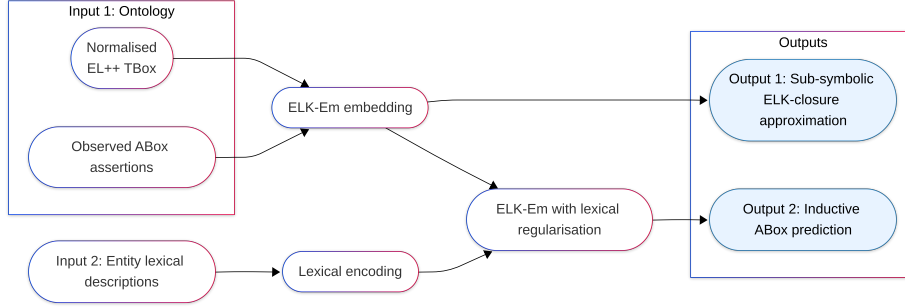


Fig. 2. Overview of the ELK-Em pipeline.

4.1 Geometric Semantics

We embed an ontology $\mathcal{O}$ in normal form over signature $(\mathbf{N}_C, \mathbf{N}_R, \mathbf{N}_I)$ into $\mathbb{R}^n$. Concepts are represented as axis-aligned boxes, individuals as points, and roles as affine transformations with bounded error over a restricted range region.

Concepts and Individuals. An axis-aligned box in $\mathbb{R}^n$ is an interval $[l, u]$ where $l, u \in \mathbb{R}^n$, interpreted as the set $\{x \in \mathbb{R}^n \mid l \leq x \leq u\}$, with inequalities taken component-wise. A box is empty if $\exists i$ such that $u_i < l_i$. In this case, we write $[l, u] = \emptyset$. Every named concept $C \in \mathbf{N}_C$ is embedded as a box $B_C = [l_C, u_C]$. Concept inclusion is interpreted as set inclusion: $B_C \subseteq B_D \Leftrightarrow l_D \leq l_C \wedge u_C \leq u_D$ (see Fig. 3(c)) and conjunction is interpreted as intersection: $B_C \cap B_D = [\max(l_C, l_D), \min(u_C, u_D)]$. Named individuals $a \in \mathbf{N}_I$ are embedded as points e_a , or equivalently as boxes $[e_a, e_a]$. A concept assertion $C(a)$ holds iff $e_a \in B_C$ (see Fig. 3(a))

Roles. Each role $r \in \mathbf{N}_R$ is represented by a triple $\eta(r) = \langle R_r, E_r, T_r \rangle$. Here $R_r = [l_r^R, u_r^R]$ is a box which represents the range of the role, $E_r = [l_r^E, u_r^E]$ is an error box, and $T_r(x) = m_r x + c_r$ is a diagonal affine map with $m_r \in \mathbb{R}_{>0}^n$ and $c_r \in \mathbb{R}^n$. Role assertions are interpreted as

$$r(a, b) \quad \text{iff} \quad e_b \in R_r \wedge e_a \in T_r(e_b) \oplus E_r,$$

where $B_x \oplus B_y$ means $[l_x + l_y, u_x + u_y]$. The error box allows multiple individuals to satisfy $r(a, b)$ for the same b (see Fig. 3(b)). This induces the semantics of existential restriction: for any concept (possibly complex), we define

$$B_{\exists r, C} = T_r(B_C \cap R_r) \oplus E_r.$$

Intuitively, $\exists r, C$ contains those points that can be related by r to a point lying in both B_C and R_r . The box R_r directly represents the range of r , whereas the domain of r is defined with existential restriction to $\top$:

$$B_{\text{ran}(r)} = R_r, \quad B_{\text{dom}(r)} = B_{\exists r, \top} = T_r(R_r) \oplus E_r.$$

Consequently, range and domain restrictions reduce to ordinary box inclusion:

$$\text{ran}(r) \sqsubseteq C \Leftrightarrow R_r \subseteq B_C, \quad \text{dom}(r) \sqsubseteq C \Leftrightarrow T_r(R_r) \oplus E_r \subseteq B_C.$$

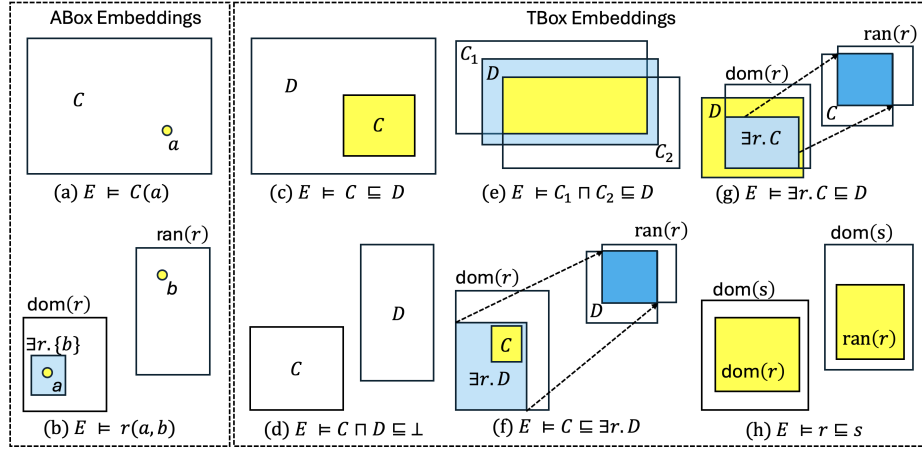


Fig. 3. Geometric semantics of ABox axioms (left) and TBox axioms (right) in $\mathcal{EL}^{++}$ under ELK-Em embeddings.

Proposition 1. For any role embedding $\eta(r)$ and concepts A, B , if $B_A \subseteq B_B$, then $B_{\exists r.A} \subseteq B_{\exists r.B}$.

Proof. Assume $B_A \subseteq B_B$. Then $B_A \cap R_r \subseteq B_B \cap R_r$. In general, affine transformations preserve inclusions, so $T_r(B_A \cap R_r) \subseteq T_r(B_B \cap R_r)$. Minkowski addition with E_r also preserves inclusion, so $T_r(B_A \cap R_r) \oplus E_r \subseteq T_r(B_B \cap R_r) \oplus E_r$. Therefore, $B_{\exists r.A} \subseteq B_{\exists r.B}$. $\square$

Role inclusion. For roles $r, s \in \mathbf{N}_R$, we define

$$\eta(r) \preceq \eta(s) \quad :\Longleftrightarrow \quad R_r \subseteq R_s \wedge \forall y \in R_r, T_r(y) \oplus E_r \subseteq T_s(y) \oplus E_s.$$

Thus, every individual in the range of r is also admissible for s , and every existential restriction of r to an individual in R_r is contained within the existential restriction of s to that individual.

Proposition 2. If $\eta(r) \preceq \eta(s)$, then:

1. $r(a, b) \Rightarrow s(a, b)$ for all individuals a, b ;
2. $B_{\exists r.C} \subseteq B_{\exists s.C}$ for every concept C .

Proof. Assume $\eta(r) \preceq \eta(s)$. For (1), let $r(a, b)$ hold. Then $e_b \in R_r$, hence $e_b \in R_s$ since $R_r \subseteq R_s$. Moreover, by the definition of $\preceq$ with $y = e_b$, we have $T_r(e_b) \oplus E_r \subseteq T_s(e_b) \oplus E_s$. Since $e_a \in T_r(e_b) \oplus E_r$, it follows that $e_a \in T_s(e_b) \oplus E_s$, hence $s(a, b)$.

For (2), let $x \in B_{\exists r.C}$. Then there exists $y \in B_C \cap R_r$ such that $x \in T_r(y) \oplus E_r$. Since $R_r \subseteq R_s$, we also have $y \in B_C \cap R_s$. Again by $\eta(r) \preceq \eta(s)$, $T_r(y) \oplus E_r \subseteq T_s(y) \oplus E_s$. Hence $x \in T_s(y) \oplus E_s$ and $x \in B_{\exists s.C}$. Thus, $B_{\exists r.C} \subseteq B_{\exists s.C}$. $\square$

Proposition 3. Role Composition. *The representation of roles is also closed under composition if defined as follows: for roles r and s , we define their composite $\eta(r \circ s) = \langle R_{r \circ s}, E_{r \circ s}, T_{r \circ s} \rangle$ with*

$$R_{r \circ s} = R_s \cap T_s^{-1}(R_r \ominus E_s), \quad E_{r \circ s} = T_r(E_s) \oplus E_r, \quad T_{r \circ s} = T_r \circ T_s$$

where T_s^{-1} is the inverse diagonal affine map of T_s and $B_x \ominus B_y = [l_x - l_y, u_x - u_y]$. Note that $R_{r \circ s}$ may be empty.

This is semantically meaningful and indicates that the role chain has an empty range box, implying an empty composite (see Appendix A.1 for a visual representation).

Proof. Let $x \in B_{\exists(r \circ s).C}$. Then $x \in T_{r \circ s}(y) \oplus E_{r \circ s}$ for some $y \in B_C \cap R_{r \circ s}$. By definition of $R_{r \circ s}$, we have $y \in B_C \cap R_s$ and $T_s(y) \oplus E_s \subseteq R_r$. Also, $T_s(y) \oplus E_s \subseteq T_s(B_C \cap R_s) \oplus E_s = B_{\exists s.C}$, hence $T_s(y) \oplus E_s \subseteq B_{\exists s.C} \cap R_r$. Since

$$x \in (T_r \circ T_s)(y) \oplus T_r(E_s) \oplus E_r = T_r(T_s(y) \oplus E_s) \oplus E_r,$$

by the definitions of $T_{r \circ s}$ and $E_{r \circ s}$, it follows that $x \in T_r(B_{\exists s.C} \cap R_r) \oplus E_r = B_{\exists r.(\exists s.C)}$. Thus $B_{\exists(r \circ s).C} \subseteq B_{\exists r.(\exists s.C)}$. $\square$

Remark. The converse inclusion fails in general. For instance, in one dimension let $B_C = [0.4, 0.6]$, $R_s = R_r = [0, 1]$, $T_s = T_r = \text{id}$, $E_s = [-0.6, 0.6]$ and $E_r = [-0.1, 0.1]$. Then $B_{\exists r.(\exists s.C)} = [-0.1, 1.1]$, whereas $R_r \ominus E_s = [0.6, 0.4] = \emptyset$, so $R_{r \circ s} = \emptyset$ and hence $B_{\exists(r \circ s).C} = \emptyset$. Therefore $B_{\exists r.(\exists s.C)} \not\subseteq (B_{\exists(r \circ s).C})$.

Proposition 4. *For roles $r, s \in \mathbf{N}_R$, $\eta(r) \preceq \eta(s)$ iff $R_r \subseteq R_s$ and*

$$T_r(y) \oplus E_r \subseteq T_s(y) \oplus E_s \quad \text{for } y \in \{l_r^R, u_r^R\}.$$

Proof. By definition, $\eta(r) \preceq \eta(s)$ means $R_r \subseteq R_s$ and

$$T_r(y) \oplus E_r \subseteq T_s(y) \oplus E_s \quad \text{for all } y \in R_r.$$

Since both lower- and upper-bound differences are affine in y , it suffices to check them at the endpoints of the box R_r , which yields the stated inequalities. $\square$

Finally, we have everything we need to define the embedding semantics.

- $E \models C(a)$ iff $e_a \in B_C$;
- $E \models r(a, b)$ iff $e_b \in R_r \wedge e_a \in T_r(e_b) \oplus E_r$;
- $E \models C \sqsubseteq D$ iff $B_C \subseteq B_D$, where C and D are possibly complex concepts;
- $E \models r \sqsubseteq s$ iff $\eta(r) \preceq \eta(s)$.

4.2 Model Training

In this section, we define training losses for TBox and ABox supervision, together with the scoring functions used at evaluation time. All losses are margin-based with $\gamma \in \mathbb{R}_{>0}^n$, and $\text{ReLU}(x) = \max(x, 0)$ is applied component-wise to vectors. We first

introduce a small set of primitive losses from which the loss functions for each normal form are constructed.

We follow previous works [14,11,29] and use distance-based loss functions. The per-dimension separation vector

$$d(B_x, B_y) = |c_x - c_y| - o_x - o_y$$

where $c = (u + l)/2$ and $o = (u - l)/2$.

Inclusion loss. For boxes B_x and B_y , inclusion $B_x \subseteq B_y$ is equivalent to $|c_x - c_y| + o_x \leq o_y$. We therefore define

$$l_{\subseteq}^{\gamma}(B_x, B_y) = \begin{cases} \|\text{ReLU}(d(B_x, B_y) + 2o_x + \gamma)\|_1, & B_y \neq \emptyset, \\ \|\text{ReLU}(o_x + \gamma)\|_1, & B_y = \emptyset. \end{cases}$$

Overlap loss. The boxes overlap iff $d(B_x, B_y) \leq 0$ component-wise. Therefore, to make two boxes B_x and B_y overlap, we use

$$l_{\cap+}^{\gamma}(B_x, B_y) = \|\text{ReLU}(d(B_x, B_y) + \gamma)\|_1$$

Disjoint loss. Two boxes are disjoint iff $\exists i$ such that $d_i(B_x, B_y) \geq 0$, so we define the disjoint loss as

$$l_{\cap-}^{\gamma}(B_x, B_y) = \text{ReLU}(\gamma - \max_i d_i(B_x, B_y)).$$

Lemma 1. For non-empty boxes B_x and B_y , with margin $\gamma \in \mathbb{R}_{>0}^n$:

- $l_{\subseteq}^{\gamma}(B_x, B_y) = 0 \implies B_x \subseteq B_y$
- $l_{\cap+}^{\gamma}(B_x, B_y) = 0 \implies B_x \cap B_y \neq \emptyset$
- $l_{\cap-}^{\gamma}(B_x, B_y) = 0 \implies B_x \cap B_y = \emptyset$

TBox Training. With these loss primitives, we define one loss function for each normal form as described in Table 2. The full TBox loss is the sum over all normalised axiom losses:

$$\mathcal{L}_{\mathcal{T}} = \sum_i \sum_{\alpha \in \mathcal{T}_{\text{NF}_i}} l_i(\alpha),$$

where l_i is the corresponding loss from Table 2.

TBox Scoring. During training, TBox axioms are optimised using the margin-based losses above. At evaluation time, we instead use the corresponding margin-free violations. For each normal-form loss l_i , let l_i^0 denote the version obtained by setting the margin to 0 in all primitive losses occurring in l_i . We then score an axiom α by

$$s(\alpha) = \exp(-l_i^0(\alpha)).$$

Thus, a satisfied axiom receives score 1, and the score decreases smoothly as the degree of violation increases. In practice, we optimise this loss over mini-batches of normalised TBox axioms.

Table 2. Loss terms for normalised $\mathcal{EL}^{++}$ axiom forms. The additional overlap terms in NFs 2,3,4 and 6 encourage non-empty interaction between the corresponding boxes and help avoid degenerate solutions. The NF5 loss is derived directly from Proposition 4.

Normal Form	Loss
NF1	$l_1(C, D) = l_{\subseteq}^{\gamma}(B_C, B_D)$
NF2	$l_2(C_1, C_2, D) = l_{\subseteq}^{\gamma}(B_{C_1 \cap C_2}, B_D) + l_{\cap+}^{\gamma}(B_{C_1}, B_{C_2})$
NF2_⊥	$l_2(C_1, C_2, \perp) = l_{\cap-}^{\gamma}(B_{C_1}, B_{C_2})$
NF3	$l_3(C, r, D) = l_{\subseteq}^{\gamma}(B_C, B_{\exists r.D}) + l_{\cap+}^{\gamma}(R_r, B_D)$
NF4	$l_4(r, C, D) = l_{\subseteq}^{\gamma}(B_{\exists r.C}, B_D) + l_{\cap+}^{\gamma}(R_r, B_C)$
NF5	$l_5(r, s) = l_{\subseteq}^{\gamma}(R_r, R_s) + \sum_{y \in \{l_r^R, u_r^R\}} l_{\subseteq}^{\gamma}(T_r(y) \oplus E_r, T_s(y) \oplus E_s)$
NF6	$l_6(r_1, r_2, s) = l_5(r_1 \circ r_2, s) + l_{\cap+}^{\gamma}(R_{r_2}, T_{r_2}^{-1}(R_{r_1} \ominus E_{r_2}))$

Negative Sampling. We follow previous works and use negative samples during training to avoid degenerate solutions. For each positive NF1 axiom $C \sqsubseteq D$, we generate $k \geq 1$ corrupted axioms by replacing D with a uniformly sampled concept from the concept vocabulary. For each positive NF3 axiom $C \sqsubseteq \exists r.D$, we generate $k \geq 1$ corrupted axioms by replacing either C or D with equal probability, while keeping r fixed. To contrast positive and corrupted axioms, we use a Pearson loss. For each positive or corrupted axiom α , we compute the corresponding margin-free score $s(\alpha) = \exp(-l_i^0(\alpha))$. Let $\mathbf{s} \in \mathbb{R}^N$ be the vector of scores and $\mathbf{y} \in \{0, 1\}^N$ the corresponding binary labels. The Pearson loss is defined as

$$\mathcal{L}_{\text{contrast}} = 1 - \frac{(\mathbf{s} - \bar{\mathbf{s}})^{\top}(\mathbf{y} - \bar{\mathbf{y}})}{\|\mathbf{s} - \bar{\mathbf{s}}\| \|\mathbf{y} - \bar{\mathbf{y}}\| + \epsilon}.$$

Minimising this loss encourages separation between positive and corrupted axioms, assigning higher scores to positives and lower scores to negatives. This provides a lenient contrastive signal without introducing an additional margin hyperparameter, while adhering to $\mathcal{EL}^{++}$ semantics. This method is only used when training exclusively with TBox data, as in section 5.1.

ABox Training. We train on concept assertions $C(a)$ using the singleton-box encoding $a \mapsto \{a\} = [e_a, e_a]$, which reduces concept membership to a special case of NF1. Since ontologies typically provide only positive assertional facts, we generate negative concept assertions by corrupting the class in positive assertions, while filtering out known training positives. Concretely, positive and corrupted concept assertions are learned with the following losses:

$$l_{C(a)} = l_1^{\gamma}(\{a\}, C), \quad l_{\neg C(a)} = \text{ReLU}(\delta - l_1^{\gamma}(\{a\}, C)),$$

where $\delta = -\log \tau$ for a target score threshold $\tau \in (0, 1)$ set as a model parameter. The full ABox loss is then

$$\mathcal{L}_{\mathcal{A}} = \sum_{C(a) \in \mathcal{A}^+} l_{C(a)} + \sum_{\neg C(a) \in \mathcal{A}^-} l_{\neg C(a)},$$

where $\mathcal{A}^+$ and $\mathcal{A}^-$ denote the sets of positive and corrupted concept assertions, respectively. The same construction can be extended to role assertions $r(a, b)$ by viewing them as singleton instances of NF3, but in this work ABox supervision is restricted to concept assertions.

4.3 Theoretical Properties

We now state the main formal properties of ELK-Em. These results show that zero geometric loss is not merely an optimisation criterion, but entails semantic satisfaction of the embedded ontology. Bourgaux et al. [4] show that, among prior geometric embedding methods for $\mathcal{EL}^{++}$, BoxEL is the only method with a soundness guarantee [28]. However, BoxEL does not cover role inclusion and composition axioms. ELK-Em extends this line of work by providing soundness guarantees to full normalised $\mathcal{EL}^{++}$.

Moreover, to the best of our knowledge, previous methods do not provide guarantees relating the learned embedding to closure under ELK’s basic inference rules. ELK-Em provides such guarantees for all rules listed below, except for consequences that essentially require role composition. Full proofs are given in Appendix A.2 and A.3.

Theorem 1. *Let $\mathcal{O}$ be a normalised ontology, and let E be an embedding such that every TBox axiom in $\mathcal{O}$ and every positive ABox assertion in $\mathcal{O}$ has zero geometric loss. Then E is a model of $\mathcal{O}$.*

Theorem 2. *Let $\mathcal{O}$ be a normalised ontology, and let E be an embedding such that every TBox axiom in $\mathcal{O}$ and every positive ABox assertion in $\mathcal{O}$ has zero geometric loss. Then the embedding satisfies the closure of $\mathcal{O}$ under the $\mathcal{EL}^{++}$ inference rules $R_0, R_{\top}, R_{\perp}, R_{\sqcap}^-, R_{\sqcap}^+, R_{\exists}, R_{\sqsubseteq}$ and R_H (see Figure 1). In general, this does not extend to consequences requiring rule $R_{\circ}$.*

4.4 Lexical Regularisation

In many prediction settings, new individuals lack ABox assertions but have informative non-logical descriptions. This is reflected in community tasks and resources: CAFA predicts GO terms for proteins using their sequence data [9]; the SNOMED CT Entity Linking Challenge assigns SNOMED CT concepts to clinical discharge summaries [8]; rare-disease resources such as CADA and PhEval support phenotype-driven prediction from clinical or phenotypic descriptions [19,5]; and MONDO integrates disease representations across sources, where diseases may be associated with names, synonyms, definitions, and textual metadata [26]. Across these settings, external descriptions provide information that can help position new individuals relative to a curated TBox.

We use lexical regularisation to transfer this non-logical information into the ontology embedding space. Let $a_1, \dots, a_m$ be a batch of individuals. For each individual a_i ,

let $z_i \in \mathbb{R}^k$ denote an external representation obtained from information available at prediction time, and let $e_i \in \mathbb{R}^n$ denote its learned geometric embedding. The external representation may be produced by a task-specific encoder, such as ESM-2 for protein sequences [15] or a Sentence-BERT-style encoder for clinical text [22].

We compute lexical similarity using cosine similarity:

$$s_{ij}^{lex} = \frac{z_i^\top z_j}{\|z_i\|_2 \|z_j\|_2}.$$

This measures the angular similarity of the external representations, making it insensitive to their magnitudes. Since $s_{ij}^{lex} \in [-1, 1]$, we rescale it to $[0, 1]$:

$$\bar{s}_{ij}^{lex} = \frac{1 + s_{ij}^{lex}}{2}.$$

Geometric similarity is defined from the squared Euclidean distance between ontology embeddings:

$$s_{ij}^{geo} = \exp\left(-\lambda \frac{\|e_i - e_j\|_2^2}{n}\right), \quad \lambda > 0.$$

This gives high similarity to nearby embeddings and decays smoothly with distance; n normalises by the embedding dimension and λ controls the decay rate.

The lexical regularisation loss aligns the pairwise similarity structure induced by the external descriptions with the geometry of the learned embeddings:

$$\mathcal{L}_{lex} = \frac{1}{\binom{m}{2}} \sum_{1 \leq i < j \leq m} (\bar{s}_{ij}^{lex} - s_{ij}^{geo})^2.$$

This term encourages individuals with similar external descriptions to be placed close together in the ontology embedding space. For individuals with ABox assertions, it acts as an auxiliary regulariser; for inductive individuals without ABox supervision, it provides a placement signal from their external descriptions. In Section 5.2, we evaluate this strategy for protein function prediction using ESM-2 sequence representations [15].

5 Evaluation

In this section, we assess ELK-Em’s ability to approximate the ELK-derived deductive closure, and then evaluate its effectiveness for inductive protein function prediction using protein-sequence encodings as the lexical regularisation input.

5.1 Deductive Closure Approximation

We evaluate how well ELK-Em approximates the deductive closure of an ontology, following the experimental setup introduced by Box²EL [11].

Datasets and baselines. We use the benchmarks and evaluation protocol introduced by [11], consisting of the biomedical ontologies GO, GALEN, and ANATOMY (Uberon).

Table 3. Results on the approximation of the deductive closure with GALEN, GO, and Anatomy ontologies. ELK-Em results are averaged over five random seeds.

	Model	H@1	H@10	H@100	Med	MRR	MR	AUC
GALEN	ELEm	0.00	0.04	0.20	1807	0.01	4405	0.81
	EmEL ⁺⁺	0.00	0.04	0.18	2049	0.01	4634	0.81
	BoxEL	0.00	0.00	0.01	6906	0.00	7925	0.67
	ELBE	0.00	0.06	0.16	1785	0.02	3974	0.84
	Box ² EL	0.01	0.09	0.24	1003	0.03	2833	0.88
	<u>ELK-Em</u>	0.00	0.06	0.31	301	0.02	2469	0.89
GO	ELEm	0.00	0.04	0.22	1629	0.02	7377	0.84
	EmEL ⁺⁺	0.00	0.04	0.19	1346	0.01	6557	0.86
	BoxEL	0.00	0.00	0.13	1085	0.00	5359	0.88
	ELBE	0.00	0.06	0.21	935	0.02	3846	0.92
	Box ² EL	0.00	0.08	0.49	107	0.04	1689	0.96
	<u>ELK-Em</u>	0.00	0.07	0.58	74	0.03	936	0.98
Anatomy	ELEm	0.00	0.07	0.28	901	0.02	7958	0.93
	EmEL ⁺⁺	0.00	0.07	0.26	1576	0.02	10976	0.90
	BoxEL	0.01	0.10	0.24	838	0.04	9156	0.92
	ELBE	0.00	0.08	0.32	336	0.03	2312	0.98
	Box ² EL	0.01	0.09	0.44	152	0.04	1599	0.99
	<u>ELK-Em</u>	0.01	0.14	0.44	139	0.05	1168	0.99

The benchmark contains class-signature sizes of 45,895 for GO, 23,143 for GALEN, and 106,258 for ANATOMY. After removing inferred subsumptions already present in the training axioms, the benchmark contains 3,309 inferred subsumptions for GO, 18,138 for GALEN, and 2,130 for ANATOMY, using a 90/10 evaluation-validation split. For each ontology, models are trained on the full normalised axiom set. The deductive closure is computed with the ELK reasoner [13]. We compare against the published results for ELEm [14], EmEL⁺⁺ [17], BoxEL [28], ELBE [20], and Box²EL [11].

Evaluation protocol. Evaluation is formulated as a ranking task. For each held-out entailed subsumption $C \sqsubseteq D$, the query concept C is fixed and the entire concept signature is ranked in descending order of subsumption score as candidate superclasses. We report Mean Rank (MR), Mean Reciprocal Rank (MRR), Median Rank (Med), AUC, and Hits@ k for $k \in \{1, 10, 100\}$ under the unfiltered protocol of [11].

Experimental protocol. During training, ELK-Em uses the negative sampling and Pearson contrastive objective defined in the previous section, with $k = 5$ corrupted axioms per positive axiom for GO, and $k = 3$ for GALEN and Anatomy. Models are trained for 10,000 epochs with learning rate 0.01 and decay by a factor of 0.1 after 5,000 epochs. The model checkpoint is selected by highest validation AUC and evaluated every 100 epochs. All reported ELK-Em results are averaged across five random seeds.

Results. Table 3 shows that ELK-Em achieves the best MR, Med, and AUC on all three ontologies, and the best or tied-best Hits@100 throughout. The main exception is strict top-rank retrieval: Box²EL obtains higher Hits@10 and MRR on GALEN and

GO. This is consistent with selecting checkpoints with AUC as the validation metric, which favours global separation over the full ranking rather than optimisation of the first few positions. These rank-based metrics should also be interpreted under the unfiltered protocol, where alternative asserted or entailed superclasses are not removed from the candidate list. Consequently, ranking another valid superclass above the held-out target is still penalised, while non-entailed candidates may be consistent rather than contradictory. Thus, the ranking space mixes false negatives, genuine non-entailments, and consistent non-consequences. Taken together, these results show that ELK-Em provides stronger global separation of entailed subsumptions across the candidate space, while remaining competitive on strict top-rank retrieval.

5.2 Protein Function Prediction

We evaluate ELK-Em with and without lexical regularisation on inductive ABox prediction for protein-function prediction.

Dataset. We use a CAFA5-derived benchmark over human proteins and Gene Ontology (GO) annotations [9]. The benchmark contains 25,125 human proteins, which we split by protein identity into 20,100 training proteins, 2,512 validation proteins, and 2,513 test proteins using an 80/10/10 split. This yields 843,165 training annotations, 107,386 validation annotations, and 113,947 test annotations. Protein annotations are obtained from a CAFA5-derived training annotation file, and only training proteins contribute observed concept assertions during model training. Validation and test proteins are held out at the individual level, so no observed protein–function assertions for these proteins are included in the training ABox.

Evaluation protocol. For a protein p and GO concept G , ELK-Em assigns the concept assertion score $s(G(p)) = \exp(-l_1^0(p, G))$, where l_1^0 is the zero-margin concept-assertion loss. At a decision threshold t , we predict all candidate GO terms whose score is at least t . Following the protein-centric CAFA evaluation protocol [12], precision and recall are computed after propagating both predictions and ground-truth annotations through the GO hierarchy. The operating threshold is selected on the validation split as the threshold attaining the maximum validation $F_1(t)$, i.e. validation $F_{\max}$. This fixed threshold is then applied unchanged to the test split, where we report precision, recall, and F_1 (see Appendix A.5 for details). For lexical regularisation, ELK-Em uses ESM-2 embeddings derived from transformer-based protein language models trained on amino-acid sequences (see Appendix A.4 for additional details).

Experimental protocol. We train ELK-Em in dimension 100, with and without lexical regularisation based on fixed 480-dimensional ESM-2 embeddings. Models are trained with Adam for 10,000 epochs at learning rate 0.01, using batches of size 1024 for both TBox and ABox axioms. The learning rate is kept constant for the first half of training and then cosine-decayed to 10^{-4} . For protein–function prediction, negative examples are generated by class corruption with $k = 2$ sampled negatives per positive assertion. All runs use random seed 42, and evaluation is performed on the final checkpoint using the threshold selected on the validation split.

Table 4. Inductive protein-function prediction on the human CAFA5 split.

Setting	Macro P	Macro R	$F_{\max}$
ELK-Em w/o lexical regularisation	0.0539	0.8439	0.1014
ELK-Em with lexical regularisation	0.2550	0.5212	0.3425

Results. Table 4 reports an ablation of lexical regularisation. Without lexical regularisation, held-out proteins have no assertional evidence and no sequence-based constraint tying them to the training proteins, so performance reflects the ability of the learned concept geometry alone to support inductive placement. In this case, validation and test protein embeddings remain close to their random initialisation, which causes many held-out proteins to be assigned broad, high-level GO terms. This leads to high recall but very low precision. With lexical regularisation, held-out proteins are positioned using sequence similarity, allowing ELK-Em to transfer functional information from nearby annotated proteins. The comparison therefore measures the contribution of sequence-derived individual placement to inductive ABox prediction.

6 Conclusions and Future Work

We presented ELK-Em, a novel geometric embedding model for $\mathcal{EL}^{++}$ ontologies. ELK-Em, unlike the state-of-the-art, (i) approximates the ELK-derived deductive closure, and (ii) supports inductive ABox inference from lexical and domain-specific similarity. The novel parametrisation of roles provides theoretical guarantees for soundness and enables the coverage of most of the ELK closure. ELK-Em also enables the placement of previously unseen individuals without requiring observed ABox assertions, making it suitable for applications such as protein-function prediction.

Empirically, experiments on GALEN, GO, and Anatomy demonstrated that ELK-Em more accurately approximates the ELK-derived deductive closure than existing geometric baselines. Furthermore, using ESM-2 protein representations, ELK-Em successfully supported inductive concept-assertion prediction on CAFA5-derived human protein-function datasets, illustrating the benefit of integrating ontology geometry with domain-specific similarity signals.

In the near future, we plan to (i) investigate the application of the lexical regularisation to the placement of concepts and roles with unseen TBox axioms, (ii) analyse more expressive representations to capture a broader fragment of OWL semantics, (iii) extend the application of the ELK-Em embeddings to other tasks such as protein-protein interaction prediction, and (iv) design evaluation metrics that more accurately assess deductive-closure approximation in geometric ontology embeddings.

Supplemental Material. Codes and resources are available under Apache-2.0 license: <https://github.com/city-artificial-intelligence/elk-em>.

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A Appendix

A.1 Role Composition

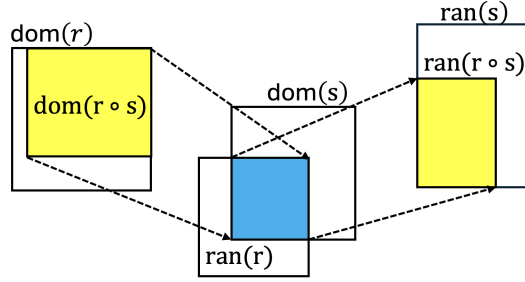


Fig. 4. Geometric semantics for role composition in $\mathcal{EL}^{++}$ under ELK-Em embeddings. If the range of r and the domain of s don't intersect, their composite is empty. This models the semantics of role composition, i.e. $(r \circ s)^{\mathcal{I}} = \{(x, z) \mid \exists y \in \Delta^{\mathcal{I}} : (x, y) \in r^{\mathcal{I}} \wedge (y, z) \in s^{\mathcal{I}}\}$.

A.2 Proof of Soundness

Theorem 1 *Let $\mathcal{O}$ be a normalised ontology, and let E be an embedding such that every TBox axiom in $\mathcal{O}$ and every positive ABox assertion in $\mathcal{O}$ has zero geometric loss. Then E is a model of $\mathcal{O}$.*

Proof. Since $\mathcal{O}$ is normalised, it suffices to check each normal form. By Lemma (1), zero inclusion, overlap, and disjointness losses imply the corresponding semantic box conditions.

- **NF1:** $C \sqsubseteq D$. If $l_1(C, D) = 0$, then $l_{\subseteq}^{\gamma}(B_C, B_D) = 0$. Hence $B_C \subseteq B_D$, so $E \models C \sqsubseteq D$.
- **NF2:** $C_1 \sqcap C_2 \sqsubseteq D$. If $l_2(C_1, C_2, D) = 0$, then $l_{\subseteq}^{\gamma}(B_{C_1 \sqcap C_2}, B_D) = 0$. Hence $B_{C_1 \sqcap C_2} \subseteq B_D$. Since $B_{C_1 \sqcap C_2} = B_{C_1} \cap B_{C_2}$, we obtain $E \models C_1 \sqcap C_2 \sqsubseteq D$. The additional overlap term is not needed for soundness, but prevents degenerate empty intersections.
- **NF2_⊥:** $C_1 \sqcap C_2 \sqsubseteq \perp$. If $l_2(C_1, C_2, \perp) = 0$, then $l_{\cap-}^{\gamma}(B_{C_1}, B_{C_2}) = 0$. Hence $B_{C_1} \cap B_{C_2} = \emptyset = B_{\perp}$, so $E \models C_1 \sqcap C_2 \sqsubseteq \perp$.
- **NF3:** $C \sqsubseteq \exists r.D$. If $l_3(C, r, D) = 0$, then $l_{\subseteq}^{\gamma}(B_C, B_{\exists r.D}) = 0$. Hence $B_C \subseteq B_{\exists r.D}$, where $B_{\exists r.D} = T_r(B_D \cap R_r) \oplus E_r$. Therefore $E \models C \sqsubseteq \exists r.D$. The overlap term $l_{\cap+}^{\gamma}(R_r, B_D)$ is not needed for soundness, but encourages a non-empty interaction between R_r and B_D .
- **NF4:** $\exists r.C \sqsubseteq D$. If $l_4(r, C, D) = 0$, then $l_{\subseteq}^{\gamma}(B_{\exists r.C}, B_D) = 0$. Hence $B_{\exists r.C} \subseteq B_D$, so $E \models \exists r.C \sqsubseteq D$. The additional overlap term again only prevents degeneracy.

- **NF5:** $r \sqsubseteq s$. If $l_5(r, s) = 0$, then $l_{\sqsubseteq}^\gamma(R_r, R_s) = 0$ and, for each $y \in \{l_r^R, u_r^R\}$, $l_{\sqsubseteq}^\gamma(T_r(y) \oplus E_r, T_s(y) \oplus E_s) = 0$. Thus $R_r \subseteq R_s$ and $T_r(y) \oplus E_r \subseteq T_s(y) \oplus E_s$ at the two endpoints of R_r . By Proposition (4), this is equivalent to $\eta(r) \preceq \eta(s)$. Hence $E \models r \sqsubseteq s$.
- **NF6:** $r_1 \circ r_2 \sqsubseteq s$. If $l_6(r_1, r_2, s) = 0$, then in particular $l_5(r_1 \circ r_2, s) = 0$. By the NF5 case applied to the composite role embedding $\eta(r_1 \circ r_2)$, we obtain $\eta(r_1 \circ r_2) \preceq \eta(s)$. Hence $E \models r_1 \circ r_2 \sqsubseteq s$. The additional overlap term in l_6 prevents the composite range from becoming trivially empty, but is not required for soundness of the role-chain inclusion.

It remains to check positive ABox assertions.

- **Concept assertions:** $C(a)$. If $l_{C(a)} = 0$, then $l_1^\gamma(\{a\}, C) = 0$. Hence $\{e_a\} \subseteq B_C$, equivalently $e_a \in B_C$. Therefore $E \models C(a)$.
- **Role assertions:** $r(a, b)$. If $l_{r(a, b)} = 0$, then $l_3^\gamma(\{a\}, r, \{b\}) = 0$. Hence $\{e_a\} \subseteq B_{\exists r, \{b\}}$. Since $B_{\exists r, \{b\}} = T_r(\{e_b\} \cap R_r) \oplus E_r$, this entails $e_b \in R_r$ and $e_a \in T_r(e_b) \oplus E_r$. Thus $E \models r(a, b)$.

Therefore every TBox axiom and every positive ABox assertion in $\mathcal{O}$ is satisfied by E . Hence E is a model of $\mathcal{O}$.

A.3 Proof of ELK-approximation

Theorem 2 *Let $\mathcal{O}$ be a normalised ontology, and let E be an embedding such that every TBox axiom in $\mathcal{O}$ and every positive ABox assertion in $\mathcal{O}$ has zero geometric loss. Then the embedding satisfies the closure of $\mathcal{O}$ under the $\mathcal{EL}^{++}$ inference rules $R_0, R_\perp, R_\top, R_\sqcap, R_\sqcup, R_\exists, R_\sqsubseteq$ and R_H (see Figure 1). In general, this does not extend to consequences requiring rule $R_\circ$.*

Proof (Proof of Theorem 2). By Theorem 1, E is a model of $\mathcal{O}$. It remains to check ELK-Em approximation of ELK's basic inference rules listed in 1.

- R_0 . This follows from reflexivity of box inclusion: $B_C \subseteq B_C$.
- $R_\top$. Since $B_\top$ is the ambient space, $B_C \subseteq B_\top$ for every concept C .
- $R_\perp$. Suppose $B_X \subseteq B_{\exists r, C}$ and $B_C \subseteq B_\perp$. Since $B_\perp = \emptyset$, this gives $B_C = \emptyset$. Hence $B_{\exists r, C} = T_r(B_C \cap R_r) \oplus E_r = \emptyset$. Therefore $B_X \subseteq B_\perp$, so $E \models X \sqsubseteq \perp$.
- $R_\sqcap$. If $B_C \subseteq B_{D_1 \sqcap D_2}$, then $B_C \subseteq B_{D_1} \cap B_{D_2}$. Hence $B_C \subseteq B_{D_1}$ and $B_C \subseteq B_{D_2}$.
- $R_\sqcup$. If $B_C \subseteq B_{D_1}$ and $B_C \subseteq B_{D_2}$, then $B_C \subseteq B_{D_1} \cap B_{D_2} = B_{D_1 \sqcap D_2}$.
- $R_\exists$. If $B_C \subseteq B_D$, then by monotonicity of existential restriction, Proposition (1) gives $B_{\exists r, C} \subseteq B_{\exists r, D}$. Hence the existential consequence is satisfied.
- $R_\sqsubseteq$. This follows from transitivity of set inclusion. If $B_C \subseteq B_D$ and $B_D \subseteq B_E$, then $B_C \subseteq B_E$.
- R_H . If $E \models r \sqsubseteq s$, then $\eta(r) \preceq \eta(s)$. By Proposition (2), $B_{\exists r, C} \subseteq B_{\exists s, C}$ for every concept C . The same proposition also gives preservation of role assertions: $r(a, b)$ implies $s(a, b)$.

Thus every consequence obtained using $R_0, R_\top, R_\perp, R_\sqcap, R_\sqcup, R_\exists, R_\sqsubseteq$, and R_H is satisfied by E .

The statement does not extend in general to consequences requiring $R_\circ$. The composite-role construction guarantees only $B_{\exists(r \circ s).C} \subseteq B_{\exists r.(\exists s.C)}$, while the converse inclusion can fail, as shown by the counterexample following Proposition (3). Hence the embedding need not satisfy all consequences that rely on the role-composition rule $R_\circ$.

Therefore E satisfies the closure of $\mathcal{O}$ under the stated $\mathcal{EL}^{++}$ rules, but not necessarily under $R_\circ$.

A.4 ESM-2 Embeddings

ESM-2 (Evolutionary Scale Modeling v2) is a family of transformer-based protein language models trained on amino-acid sequences, the primary sequence representation of proteins [15]. Let $\mathcal{A}$ denote the amino-acid alphabet. Given a protein sequence

$$p = (a_1, \dots, a_L), \quad a_i \in \mathcal{A},$$

ESM-2 defines a mapping

$$\text{ESM2}(p) = (h_1, \dots, h_L),$$

where each $h_i \in \mathbb{R}^d$ is a contextual embedding of residue a_i , computed with respect to the full sequence p . Residue embeddings can be pooled to obtain a fixed-dimensional representation of the protein sequence. In this work, we use mean pooling:

$$z_p = \frac{1}{L} \sum_{i=1}^L h_i.$$

Given two proteins p and q , we define their ESM-2 similarity by cosine similarity between the pooled sequence embeddings:

$$s_{pq}^{\text{ESM}} = \frac{z_p^\top z_q}{\|z_p\| \|z_q\|}.$$

This gives a sequence-derived similarity signal between protein individuals. We use this signal to regularise the placement of proteins in the ontology embedding space for inductive ABox prediction.

A.5 Protein-Function Prediction Metrics

We use the protein-centric evaluation metrics for protein function prediction defined by [12]. Specifically, we report macro precision and recall over evaluated proteins, selecting the operating threshold on the validation set by maximising $F_{\max}$, and report test-set precision, recall and F_1 at this validation selected threshold.

For each target protein p , ELK-Em assigns a score to every candidate concept assertion $G(p)$, where G is a GO concept. At a decision threshold t , the predicted annotation set is

$$P_t(p) = \{G \in \mathcal{C} \mid s(G(p)) \geq t\},$$

where $\mathcal{C}$ is the set of candidate GO concepts. Evaluation is carried out with respect to the GO hierarchy for both predictions and ground-truth annotations. We write $T(p)$ for the corresponding ground-truth annotation set of protein p .

Protein-centric precision and recall at threshold t are defined as

$$\text{pr}_p(t) = \frac{|P_t(p) \cap T(p)|}{|P_t(p)|}, \quad \text{rc}_p(t) = \frac{|P_t(p) \cap T(p)|}{|T(p)|}.$$

When $P_t(p) = \emptyset$, precision is undefined and the protein is omitted from the precision average. Recall is averaged over all evaluated proteins. Macro precision and macro recall are therefore

$$\text{Pr}(t) = \frac{1}{m(t)} \sum_{p: |P_t(p)| > 0} \text{pr}_p(t), \quad \text{Rc}(t) = \frac{1}{n} \sum_{p=1}^n \text{rc}_p(t),$$

where n is the number of evaluated proteins and $m(t)$ is the number of proteins with at least one prediction at threshold t . The protein-centric F_1 score at threshold t is

$$F_1(t) = \frac{2 \text{Pr}(t) \text{Rc}(t)}{\text{Pr}(t) + \text{Rc}(t)}.$$

The maximum protein-centric F -measure is

$$F_{\max} = \max_t F_1(t).$$

In our implementation, thresholds are swept over 71 values linearly spaced between 1.0 and 0.3. We select the threshold that maximises $F_1(t)$ on the validation split and then apply this threshold unchanged on the test split. Test-set precision, recall, and F_1 are reported at this validation-selected operating point.